EXPERIMENT 17

PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE WHITE GAUSSIAN NOISE (AWGN) USING MATLAB

OBJECTIVE:

- To generate a digital communication signal in MATLAB and transmit it through an Additive White Gaussian Noise (AWGN) channel.
- To obtain and analyze the eye diagram of the received digital signal for different noise conditions.
- To measure eye-opening parameters such as eye height, eye width, crossing level and timing variation.
- To study the effect of decreasing SNR / Eb/N0 on inter-symbol interference, noise margin and timing performance.
- To compare the transmitted and received signal characteristics and evaluate communication-system performance.

PROGRAM:

```
clc;

clear;

close all;

%% Parameters

N = 100; % Number of bits

sps = 20; % Samples per symbol

SNR = 10; % Signal-to-Noise Ratio (dB)

%% Generate Random Binary Data

data = randi([0 1],1,N);

%% NRZ Encoding (0 -> 0, 1 -> 1)

tx_signal = repelem(data,sps);

%% Pass Signal through AWGN Channel

rx_signal = awgn(tx_signal,SNR,'measured');

%% Time Axis

t = (0:length(tx_signal)-1)/sps;
```

```
%% Plot Results
figure;
% Original Binary Data
subplot(4,1,1)
stairs(data,'LineWidth',1.5)
grid on
title('Original Binary Data')
xlabel('Bit Number')
ylabel('Bit')
ylim([-0.2 1.2])
% Transmitted NRZ Signal
subplot(4,1,2)
plot(t,tx_signal,'LineWidth',1.5)
grid on
title('Transmitted NRZ Signal')
xlabel('Time')
ylabel('Amplitude')
ylim([-0.5 1.5])
% Received Signal after AWGN
subplot(4,1,3)
plot(t,rx_signal,'r')
grid on
title(['Received Signal with AWGN (SNR = ',num2str(SNR),' dB)'])
xlabel('Time')
ylabel('Amplitude')
% Eye Diagram
subplot(4,1,4)
eyediagram(rx_signal,2*sps)
```

title('Eye Diagram of Received Signal')

OUTPUT:

